

# Trust-Calibrated Multi-Agent Scientific Deliberation for Mitigating Sycophantic Consensus in LLM Reasoning

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## Abstract

Large language models have done surprisingly well at answering many scientific questions. However, large language models also continue to "hallucinate" and reason incorrectly. One potential solution to these problems is multi-agent debate. Although a correct minority may exist, a confident but incorrect majority can drag this correct minority agent down and lead the group to select the wrong answer. This is known as the "sycophantic consensus." Currently, there is no mechanism within any debate system to check an individual agent's assertions against outside scientific evidence before allowing them to gain influence in the debate. Instead of using a simple vote-weighted approach, we intend to assign a trust score to every participating agent based on how much scientific evidence supports their responses. Every time agents respond during a round, the system breaks down their responses into small units of factually verifiable statements ("claims"). After breaking down the response, the system locates all relevant scientific literature that could support or contradict each claim. Because unsupported claims eventually lose influence and because the trust scores are kept in bounds, the probability increases that a correct minority will survive long enough to ultimately win. Our proposed framework produced fewer hallucinated answers than competing frameworks because all surviving claims are linked back to their originating passages of scientific evidence. To test this hypothesis, we compare our proposed framework to majority voting and other baseline systems. Once all models have finished responding to a given question, the system returns both the selected model and supporting citations for that model. The performance of our proposed framework is evaluated via four metrics: answer accuracy, consensus collapse, minority preservation, and evidence calibration.

## 1. Introduction

Large language models can answer scientific questions at a level that makes them useful for tutoring and decision support. A model can still be confidently wrong, and it can state a wrong answer in fluent language that hides the error. Multi-agent debate reduces this risk because three agents answer the same question and review each other's reasoning over rounds [1]. Chain-of-thought prompting lets each agent expose the steps behind its answer, so peers can inspect the reasoning [2]. The weak point sits at the end of the debate. Most systems combine the answers by majority vote, where every agent counts the same and the most frequent answer wins [3, 4]. A confident and wrong majority can then push a correct minority agent to drop its answer, and the vote records the wrong result as agreement. This failure is called sycophantic consensus [5, 6]. No current debate system checks an agent's claims against outside scientific evidence before that agent earns influence. This study builds a trust-calibrated multi-agent deliberation framework to close that gap. Each agent's answer is split into small factual claims. The claims are checked against retrieved scientific literature, and every agent holds a bounded trust score that rises with supporting evidence and falls with contradicting evidence. The final answer comes from trust-weighted aggregation instead of a head count. The method is tested on scientific reasoning datasets such as GPQA and MMLU-Pro, and it needs no fine-tuning because all models run as served inference.

Du et al. show that debate improves factuality and reasoning over a single pass, but their agents are treated equally and their claims are never checked against external sources [1]. Pitre et al. report that the correct answer is present in the discussion yet ignored in more than 20 percent of wrong-answer cases, and their prompt-rewriting fix still leaves the final decision on self-reported confidence [5]. The uncertainty route of Yoffe et al. reweights agent influence during the

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debate, and their own ground-truth experiment shows the gain is capped by the quality of that self-reported signal [7]. Token cost is the target of Fan et al., who decide when a debate is worth running, yet majority voting remains once the debate starts [8]. Wang et al. combine many models through static aggregation and score well on open benchmarks, with no per-agent trust and no evidence check [9]. He et al. detect a suppressed correct minority after the debate and overturn the majority, which comes too late to protect the minority while it argues [6].

Across these systems, influence is set by a vote, a self-reported score, a prompt rewrite, or a post-hoc flip, and none of them verifies an agent’s claims against evidence while the debate is running. Self-reported confidence is a weak basis because training that makes models agreeable also makes them more sycophantic and less accurate [10], and alignment training has been shown to increase agreement with wrong user positions [11]. Theory also predicts that correlated agents drift to a wrong majority unless an outside correction enters the loop [12], and multi-agent systems fail in recurring ways that diagnosis alone does not fix [13]. This study resolves the unsolved part by tying each agent’s influence to external evidence during the debate. Claims are checked against retrieved passages from scientific sources [14, 15], a step in line with work that grounds correction in external tools [16]. The trust score is bounded, so no agent is silenced or made all-powerful, and the final decision uses those weights. Scientific question sets are a suitable testbed because they carry verifiable answers and published literature that retrieval can reach, and GPQA and MMLU-Pro allow comparison against established baselines. The expected outcome is a reproducible framework that lowers consensus collapse, preserves a correct minority more often, keeps answer accuracy stable, and reports a calibrated trust signal.

Yao et al. define and measure sycophancy between debating agents, where an agent copies a peer answer instead of testing it, and they leave correction to future work [17]. Sharma et al. trace the behavior to training, because models learn to agree with plausible positions during preference tuning [18]. A survey of the field lists workflow patterns and open challenges for multi-agent systems and names reliability under social pressure as an unresolved item [19]. The field also reaches applied settings, such as debate with external knowledge, adaptive debate for educational reasoning, debate for research ideation, debate for scientific code generation, and debate in communications engineering [20, 21, 22, 23, 24]. Knowledge integration supplies the external signal that debate lacks, and a survey of LLM and knowledge-base methods reports factual gains from grounded retrieval [25].

The mitigation lines that exist each leave part of the gap open. Prompt refinement rewrites the debate prompt between rounds, and the final decision still rests on self-reported confidence [5]. Uncertainty weighting reweights agents from token-level signals, and its own ground-truth probe shows that the gain is bounded by the quality of that internal signal [7]. Confidence-weighted consensus pools diverse models with self-estimated confidence, which is the same weak channel [26]. A post-hoc classifier flips a suppressed majority after the debate ends, once the pressure on the minority has already been applied [6]. Diversity and training-time methods change how agents reason or how they are adapted, not how much each agent counts in the final vote [27, 28]. Intervention testing reports that most fixes for sycophancy are not sufficient on their own [29]. Efficiency work lowers the token cost of a debate and leaves the aggregation step unchanged [8, 30]. Across these lines, no method verifies an agent’s claims against external evidence while the debate runs, and no method uses that verification to set the weight of the agent in the final answer.

## 2. Conventional Method

### 2.1. Preliminaries

Large language models can answer many scientific questions, but they still produce confident errors. Chain-of-thought prompting makes the reasoning path visible, so a reader can follow the steps that produced an answer [2, 31]. Multi-agent debate extends this idea. Several agents answer the same question, read the reasoning of the other agents, and revise their positions over rounds before a final answer is selected [1]. The combination step is as important as the reasoning itself. Most debate systems combine the answers with a majority vote, and every agent carries the same weight in that vote [3]. A majority vote counts repeated answers. It does not check whether an answer is correct. A confident but wrong majority can push a correct minority agent to change its answer, and the vote records the wrong result as agreement. This behavior is called sycophantic consensus [5]. Debate systems usually rely on two confidence signals, namely self-reported confidence and token-level uncertainty. Neither signal checks the truth of the claim that carries it [7]. Trust in a debate answer therefore has to come from evidence outside the debate.

## 2.2. Existing Methods

Existing research lines address parts of this problem. Du et al. show that debate improves factuality and reasoning over a single pass, but their agents are treated equally and their claims are never checked against external sources [1]. Fan et al. reduce the cost of debate by predicting when a debate is worth running, and the aggregation step remains a majority vote once the debate starts [8]. Wang et al. combine model outputs through a layered pipeline and report strong results on open benchmarks, with no per-agent trust and no evidence check [9]. Zhou et al. choose specialized reasoners for subtasks through a reward-driven process, which selects who works on a task but not how much each answer counts [32].

A second group of methods targets sycophancy directly. Pitre et al. report that the correct answer is present in the discussion but ignored in more than 20 percent of wrong-answer cases. Their framework detects stalling or answer copying, rewrites the task prompt, and restarts the debate, while the final vote still uses self-reported confidence [5]. Chen et al. reduce sycophancy through preference alignment at training time, which operates on one model and does not cover a live debate [11]. Ibrahim et al. show that training models to sound warm lowers factual accuracy and raises agreement with false user beliefs [10]. He et al. detect a suppressed correct minority after the debate and flip the majority decision, which comes too late to protect the minority during the rounds [6].

A third group changes the information that agents receive. Yoffe et al. reweight agent influence during the debate with token-level uncertainty, and their ground-truth experiment shows that the gain is limited by the quality of that internal signal [7]. Gou et al. ground correction in external tools for a single agent [16]. Kushwaha and Madhavan detect contradictions between retrieved documents and score source credibility before generation, without a debate between agents [15]. Karpukhin et al. provide the dense passage retrieval method that this study adopts for its retrieval stage [14].

Theory and diagnosis complete the picture. Estornell and Liu prove that similar responses cause majority tyranny and that shared misconceptions lower accuracy as more agents share them [12]. Choi et al. separate debate from voting and show that majority voting explains a large part of the reported gain [4]. Cui et al. remove the consensus requirement and score the debate trajectory instead [33]. Pitre et al. evaluate debate process quality when ground truth is unavailable [34]. Cemri et al. build a failure taxonomy for multi-agent systems and report failure rates between 41 and 86.7 percent across seven systems, and they leave correction to future work [13]. Cau et al. show that selective persuasion drives agreement, and Yeow et al. show that feedback between agents can spread early errors [35, 36].

## 2.3. Research Gap

Three issues remain open in multi-agent debate. The first issue is the cost of a debate. iMAD predicts whether a debate is needed before it starts, which lowers the token cost on easy questions [8]. The second issue is how to detect agreement that comes from conformity or sycophancy. CONSENSAGENT detects stalling and answer copying, and it rewrites the prompt before the next round [5]. The third issue is how to decide which agent should carry more influence when the agents disagree. DebUnc weights agent influence with self-reported uncertainty, and its own diagnostic ground-truth experiment shows that the weighting mechanism works only as well as the uncertainty signal it receives [7].

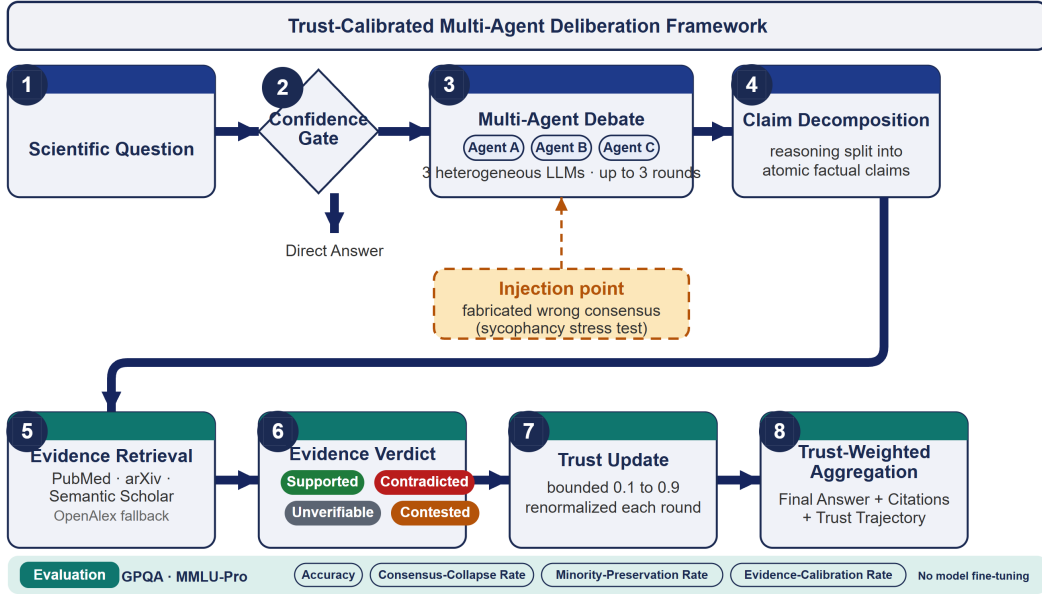
Existing fixes do not close the third issue. Static aggregation treats every proposer the same and leaves the synthesis to one model, so a confidently wrong proposer enters the final answer on equal footing [9]. Process-level diagnosis measures the health of a debate but does not change the decision [34]. A post-hoc flip protects a suppressed minority after the debate ends, when the pressure on that minority has already happened [6]. Tool-based correction verifies one agent’s output in isolation, and it does not arbitrate between agents [16]. Retrieval-side credibility scoring ranks sources before a single generation pass, with no revision rounds and no agent trust [15]. Theory predicts that correlated agents drift to the majority regardless of correctness and that shared misconceptions lower accuracy as more agents share them [12]. Training-time work shows that alignment can make this tendency stronger, because models trained to sound agreeable grow more sycophantic and less accurate [10, 11]. Empirical studies of failure also show that these failures are recurring and classifiable, and they leave correction to future work [13]. Debate-only baselines confirm that voting carries much of the reported gain, and voting is the component that cannot separate a correct answer from a repeated one [4].

Across the reviewed work, influence is set by a vote, a self-reported score, a prompt rewrite, or a post-hoc flip. No method verifies an agent’s claims against external evidence while the debate is running, and no method uses such a verification to set the weight of that agent in the final decision. This study fills that gap with a bounded trust score that is updated from retrieved evidence during the debate and used for the final aggregation.

### 3. Proposed Method

#### 3.1. Framework Overview

The framework is a deliberation layer that sits between a scientific question and its final answer. A question first passes the confidence gate. The gate chooses between a direct answer and a full debate. Inside a debate, three heterogeneous agents answer the question independently and tag the claims in their reasoning. Each claim is checked against retrieved scientific evidence. The verdicts update the trust score of every agent, and the agents revise their positions with the peer answers and their own trust standing in view. The loop runs for at most three rounds. The final answer is selected by a trust-weighted aggregation over the agent positions. The result package contains the answer, the supporting citations, the trust trajectory, and the reasoning trace of each agent. Figure 1 shows the end-to-end flow, and Figure 2 shows the system boundary and the external actors.



**Figure 1:** End-to-end flow of the trust-calibrated multi-agent deliberation framework: debate, claim-level evidence verification, per-round trust updates, and trust-weighted aggregation.

#### 3.2. Confidence Gate

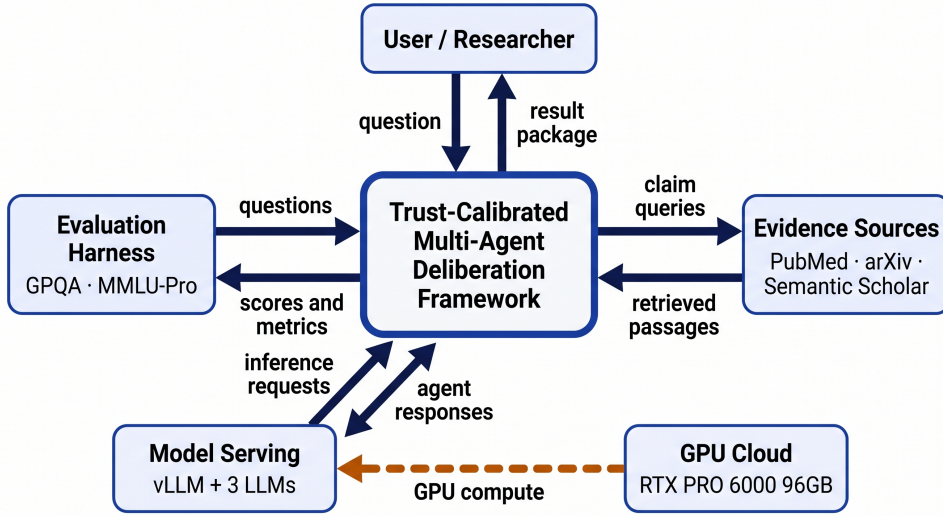
The gate keeps the cost low on easy questions. One agent answers the question with a short chain of thought and reports a confidence value, or three quick agent calls answer the same question and the gate compares the answers. A low confidence or a disagreement sends the question to the full pipeline. A high confidence sends the question to a direct answer. The gate affects throughput and not the core evaluation, because the evaluation set keeps only questions that produce divergent initial answers.

#### 3.3. Heterogeneous Debate Agents

Three agents answer the question in round zero. Each agent is a different model family, so the agents do not share the same training data and the same blind spots [12]. The development configuration uses Qwen3.5-9B, Gemma 4 12B, and Ministral-3-14B-Instruct-2512. The final configuration replaces them with Qwen3.6-27B, Gemma 4 26B A4B, and Mistral-Small-3.2-24B-Instruct-2506. All models are served through vLLM behind an OpenAI-compatible interface, and no model is fine-tuned. Each agent produces an answer, a reasoning trace, and a list of tagged claims in the form `<claim id="c1">`. The tags make claim-level checking possible in later stages.

#### 3.4. Claim Decomposition

A compound sentence can carry more than one fact. The decomposition step splits each agent answer into small factual statements, and every statement keeps the identifier of its source agent. A schema check validates the tagged



**Figure 2:** Context diagram of the framework. The evaluation harness submits questions, the evidence sources return passages, the model server runs the three agents, and the user receives the result package.

output. A failed check triggers a fallback extraction pass that recovers the claims from the raw text. A claim that no source can verify is marked unverifiable and is excluded from the trust update, so an abstention neither rewards nor penalizes the agent.

### 3.5. Source-Partitioned Retrieval and Evidence Scoring

The three agents do not share one retrieval source. Agent one sends its claims to PubMed, agent two to arXiv, and agent three to the Semantic Scholar Graph API. OpenAlex is the fallback when a source is unavailable, and retrieved results are cached so repeated runs do not hit rate limits. A cross-encoder reranker scores each claim against the retrieved passages and keeps the highest scoring passages [14]. Every claim then receives one of four verdicts, namely supported, contradicted, unverifiable, or contested. A claim with conflicting high-score passages is marked contested and reported separately in the evaluation. Figure 3 shows one claim checked against a retrieved passage.

### 3.6. Trust Update

The trust update converts the verdicts into a bounded influence score. Let  $S_i(t)$  be the raw score of agent  $i$  at round  $t$ . The raw score grows with the number of supported claims  $V_i$  and falls with the number of contradicted claims  $H_i$ :

$$S_i(t+1) = S_i(t) + \alpha V_i - \beta H_i \quad (1)$$

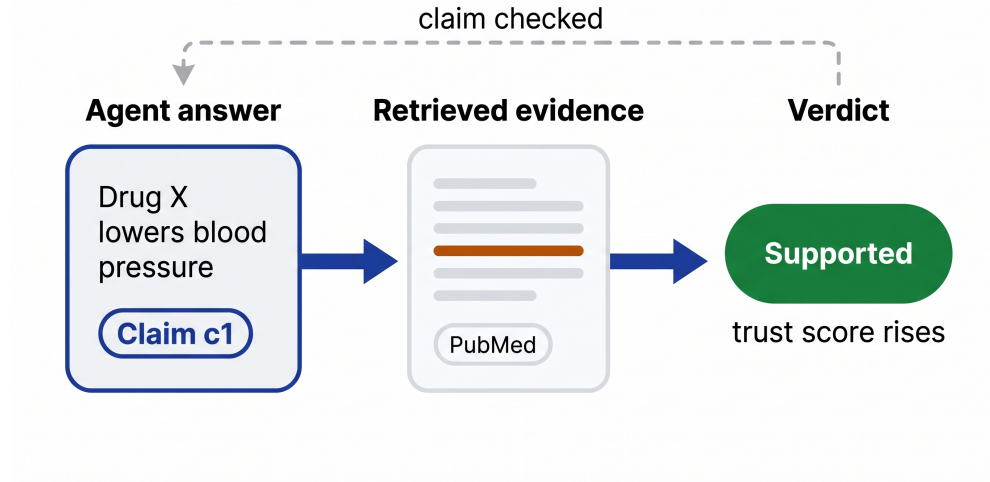
where  $\alpha$  and  $\beta$  control the gain and the loss. The raw scores are then converted into weights. A softmax maps the raw scores into positive values, a clamp keeps each value inside the interval  $[0.1, 0.9]$ , and a renormalization returns the weights to the simplex:

$$T_i = \frac{\text{clamp}(\text{softmax}(S)_i, 0.1, 0.9)}{\sum_{j=1}^N \text{clamp}(\text{softmax}(S)_j, 0.1, 0.9)} \quad (2)$$

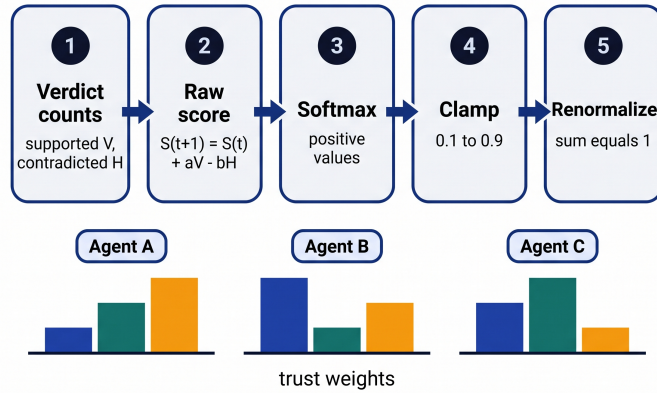
The clamp prevents two failure cases. No agent is silenced by a zero weight, and no agent dominates the debate with a weight of one. The renormalization keeps the sum of the weights at one, so the final decision stays a weighted choice over the agent positions. Equations (1) and (2) form the bounded trust update, and the whole update is a pure function of the verdict counts. Figure 4 traces one update from the verdict counts to the final weights of the three agents.

### 3.7. Deliberation Protocol

Each round follows the same order, namely positions, decomposition, retrieval, verdicts, and trust update. Table 1 lists the stages of one debated question and the number of model calls per stage. In round two and round three the

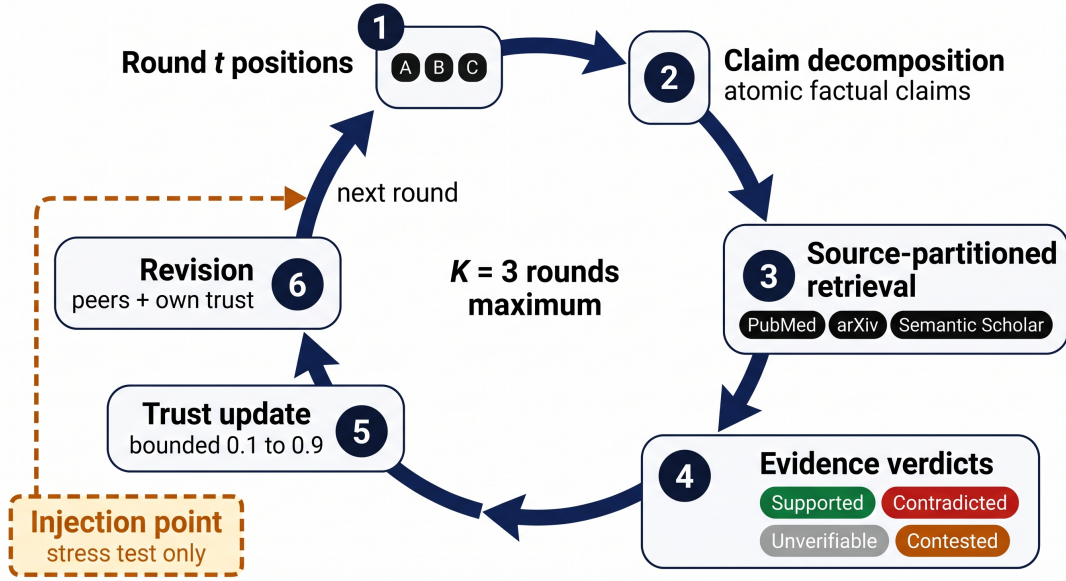


**Figure 3:** Worked example of a single claim verified against a retrieved passage. The claim is marked supported, and the trust score of the agent rises for that round.



**Figure 4:** The five steps of one trust update. The verdict counts feed the raw score, and the softmax, clamp, and renormalization steps turn that score into the final trust weights of the three agents.

revision prompt shows the agent the positions of the other agents and its own current trust value. The debate stops after round three. An injection point sits between round one and round two. At this point the evaluation harness can insert a fabricated expert consensus that contradicts a correct minority. The injection is used only in the stress tests and never in normal operation. Figure 5 shows the loop and where the injection enters it. A timeout or a parse failure marks the round inconclusive for that agent, and the debate continues with the remaining agents. Figure 6 shows the level-one data flow, including the three data stores that hold the debate state, the evidence log, and the results log.



**Figure 5:** The debate round loop. The six stages run in order for at most three rounds, and the injection point enters between round one and round two in the stress tests.

Table 1

### Stages of one debated question and their model calls.

| Stage                         | What happens                                                  | Model calls |
|-------------------------------|---------------------------------------------------------------|-------------|
| Confidence gate               | Decides between a direct answer and a full debate             | 1 or 3      |
| Round 1 positions             | Three agents answer independently and tag their claims        | 3           |
| Claim check (each round)      | Decomposition, retrieval, and verdicts label every claim      | 0           |
| Trust update (each round)     | Verdict counts move the raw score, then the weights           | 0           |
| Injection (stress tests only) | Fabricated wrong consensus enters between round 1 and round 2 | 0           |
| Round 2 revision              | Agents see peer positions and their own trust, then revise    | 3           |
| Round 3 revision              | Final revision round, after which the debate stops            | 3           |
| Aggregation                   | Trust-weighted answer is selected and packaged                | 0           |
| <b>Total</b>                  |                                                               | <b>10</b>   |

### 3.8. Trust-Weighted Aggregation

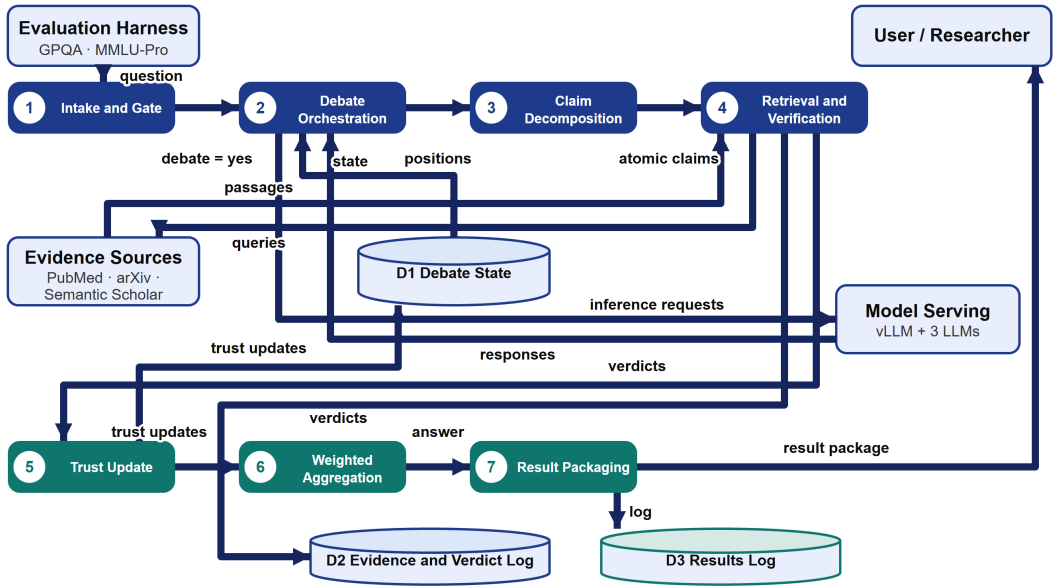
The final answer is the position with the highest weighted support:

$$a^* = \arg \max_a \sum_{i: p_i=a} T_i \quad (3)$$

where  $p_i$  is the position of agent  $i$  at the last round and  $T_i$  is its final trust weight. The aggregation ignores the plain head count. This is the step that allows a correct, evidence-backed minority to outweigh a wrong majority. The result package records the final answer, the evidence citations, the trust trajectory of every round, and the reasoning trace of every agent. The trajectories are stored because the evaluation uses them for the calibration analysis.

### 3.9. Complexity

A debated question uses about ten model calls. The gate uses one call, the initial round uses three calls, and the two revision rounds use three calls each. The retrieval traffic scales with the number of claims, and the cache removes repeated queries. The pipeline runs on a single rented GPU and requires no fine-tuning.



**Figure 6:** Level-one data flow of the framework with the debate state, the evidence log, and the results log.

## 4. Experiment

This section defines the datasets, the baselines, the metrics, the injection protocol, the serving setup, and the statistical tests that make up the evaluation. Table 2 lists the question sets.

Table 2: Datasets used in the evaluation. BrokenMath and BrokenArXiv create a false majority, HLE sets the difficulty ceiling, and GPQA and MMLU-Pro measure accuracy under stable conditions.

| Dataset      | Role               | Source                                  | License          | Size                  |
|--------------|--------------------|-----------------------------------------|------------------|-----------------------|
| BrokenMath   | Adversarial        | INSAIT-Institute, Hugging Face [37]     | CC BY-NC-SA 4.0  | 451 problems          |
| BrokenArXiv  | Adversarial        | MathArena, Hugging Face [38]            | CC BY-SA 4.0     | 87 problems           |
| HLE          | Difficulty ceiling | Center for AI Safety, Hugging Face [39] | MIT, gated       | 2,500 questions       |
| GPQA Diamond | Stable baseline    | Idavidrein, Hugging Face [40]           | CC BY 4.0, gated | 198 questions         |
| MMLU-Pro     | Stable baseline    | TIGER-Lab, Hugging Face [41]            | MIT              | Over 12,000 questions |

### 4.1. Datasets

The evaluation uses five public scientific question sets. The sets fall into three roles. Two adversarial sets create a false majority on purpose, one set holds expert-level questions that act as a difficulty ceiling, and two sets measure accuracy without adversarial pressure. Table 2 lists the role, the source, the license, and the official size of each set.

The adversarial pair is BrokenMath [37] and BrokenArXiv [38]. BrokenMath is built from 2025 competition problems that a language model perturbs into false statements, and human experts then review the perturbed versions. The release contains 451 benchmark problems written as proof tasks. BrokenArXiv turns false statements from recent arXiv papers into problems and publishes new snapshots each month, and the 2026 release used here contains 87 problems. Both sets place an agent in front of a wrong premise that reads as plausible, which is the condition that drives sycophantic agreement in a debate.

Humanity’s Last Exam [39] sets the ceiling. The set contains 2,500 expert-written questions across dozens of subject areas, and the release is gated behind a free click-through agreement with a no-redistribution rule. GPQA [40] and MMLU-Pro [41] give the stable comparison. GPQA contains 448 graduate-level multiple-choice questions in biology, physics, and chemistry, and this study uses the 198-question Diamond subset. MMLU-Pro raises the number of answer options from four to ten and adds reasoning-focused items, and its release holds over 12,000 questions across 14 domains. The two stable sets answer a different question than the adversarial pair. They check that the mechanism keeps baseline accuracy, while BrokenMath and BrokenArXiv check the behavior under pressure.

Two filters run before a question enters the pool. The divergent filter keeps only questions where the three agents give at least two different answers in round one, which removes items that the agents already agree on. Retention on the adversarial sets is about 60 to 70 percent. The answer-type filter keeps only questions with a checkable answer, such as a multiple-choice option or a number, because the injected consensus and the collapse score both need a discrete answer. The main matrix draws 1,000 questions per dataset and the pilot draws 50, and the sample caps are recorded in the project configuration.

### 4.2. Baselines

The evaluation compares the trust-calibrated system with nine baseline conditions. Baseline B1 answers with one chain-of-thought pass [2]. Baseline B2 adds the retrieval channel to the same single-agent pass, so the pair B1 and B2 isolates the effect of evidence. Baseline B3 is the vanilla debate loop of Du et al. [1], and the same loop is the Gate 0 reproduction target. Baseline B4 combines the debate loop with retrieval. Baseline B5 samples several reasoning paths and keeps the most consistent answer [42]. Baseline B6 is the layered mixture-of-agents pipeline of Wang et al. [9]. Baseline B7 sends each question to a large commercial model and records the accuracy ceiling. Baseline B8 is the trust-calibrated system. Baseline B9 is the iMAD reimplementation of Fan et al. [8], and baseline B10 is the ConsensAgent comparison of Pitre et al. [5].

The set answers the two objections that matter for this study. B5 covers the self-consistency question, and B6, B9, and B10 cover the question of an existing aggregation or efficiency method. The two retrieval pairs also serve the ablation plan. The only difference between B1 and B2, or between B3 and B4, is the evidence channel, so the comparison separates the effect of retrieval from the effect of debate.

### 4.3. Metrics

Four measures score a run, namely answer accuracy, consensus collapse rate, minority preservation rate, and evidence calibration rate. The collapse rate is the primary metric because it operationalizes the failure that the framework targets.

Let a debate carry the correct answer in the minority before the injection. The exposed set  $E$  holds the agents that answered correctly at round one and received the injected consensus. The collapse set  $C$  holds the exposed agents that change their answer in a later round while no new evidence enters the debate. The collapse rate is the share of exposed agents that drop the correct answer:

$$\text{CCR} = \frac{|C|}{|E|}. \quad (4)$$

The preservation rate is measured at the level of the debate. Let  $D$  be the set of divergent debates, namely debates with at least two distinct initial answers, and let  $P$  contain the debates where the correct answer survives to the final round, either in the position of at least one agent or in the weighted aggregate. Then

$$\text{MPR} = \frac{|P|}{|D|}. \quad (5)$$

The calibration rate checks that the trust weight means what it claims. The final trust weight of an agent is read as a confidence value, and the value is compared with the observed accuracy of the agents in the same confidence bin. Let  $B$  be the number of bins,  $n_b$  the number of agents in bin  $b$ ,  $\bar{t}_b$  the mean trust weight in that bin, and  $\text{acc}(b)$  the share of correct agents in that bin. The expected calibration error and the calibration rate are

$$\text{ECE} = \sum_{b=1}^B \frac{n_b}{N} |\text{acc}(b) - \bar{t}_b|, \quad \text{ECR} = 1 - \text{ECE}, \quad (6)$$

where  $N$  is the number of scored agents. A value of one means the trust weight matches the correctness rate in every bin. The binning follows the calibration method of Guo et al. [43]. Answer accuracy on the stable sets is the share of questions where the final answer matches the gold answer.

### 4.4. Injection Protocol

The collapse metric needs controlled pressure, and the injection protocol supplies it in six steps. The first step runs the three agents once and keeps the questions with divergent initial answers. The second step removes questions whose answers are not checkable. The third step builds the fabricated consensus from the majority answer, which is the wrong side on the questions that passed the first step. The fourth step delivers one system message that states the fabricated expert consensus. The message reaches only the agents that disagree with it, namely the correct minority, and it enters between round one and round two. No further pressure is applied after that point. The fifth step lets the debate finish its three rounds under the normal rules. The sixth step scores the abandonment with Equation (4).

The protocol is validated before the pilot runs. Two annotators label a sample of 40 questions as eligible or not eligible, and the protocol passes when the agreement reaches a Cohen’s kappa of 0.75 or higher. The kappa statistic is

$$\kappa = \frac{p_o - p_e}{1 - p_e}, \quad (7)$$

where  $p_o$  is the observed agreement and  $p_e$  is the agreement expected by chance. A kappa below the threshold sends the labeling rubric back for revision, and the pilot waits.

## 4.5. Experimental Setup

All models run as served inference, and no model is fine-tuned. The development stack uses Qwen3.5-9B, Gemma 4 12B, and Ministral-3-14B-Instruct-2512, and the final stack replaces them with Qwen3.6-27B, Gemma 4 26B A4B, and Mistral-Small-3.2-24B-Instruct-2506. The development stack runs on one rented RTX A6000 with 48 GB of memory. The checkpoints run at 4-bit precision, namely AWQ for Qwen and Ministral and QAT for Gemma, because the official Ministral FP8 checkpoint does not load on this card. The final stack runs on an RTX PRO 6000 Blackwell with 96 GB of memory.

Serving uses vLLM behind an OpenAI-compatible interface, with one server per agent and three servers on one card. Each server starts in sequence, and a readiness check gates the next start so that the memory profiling of one server does not disturb the others. A debate uses at most three rounds, the sampling temperature is 0.7, and each response is capped at 1,024 tokens. Every condition runs with three seeds, and each agent receives a seed derived from the run seed. A pilot run of ten GPQA questions on the development stack completed all three rounds for every debate, with a mean debate time of 370.8 seconds.

## 4.6. Statistical Analysis

Each condition runs with three seeds, and every reported rate carries a 95 percent confidence interval from a paired bootstrap over questions [44]. The bootstrap resamples the paired question set 10,000 times and takes the 2.5th and 97.5th percentiles of the resampled statistic. Two conditions are compared on the same resampled questions, so the interval reflects the paired design.

The effect size uses Cohen’s  $d$  on the per-question outcomes:

$$d = \frac{\bar{x}_1 - \bar{x}_2}{s_p}, \quad (8)$$

where  $\bar{x}_1$  and  $\bar{x}_2$  are the mean per-question outcomes of the two conditions and  $s_p$  is the pooled standard deviation. A condition counts as an improvement only when the confidence interval of the difference stays above zero across the three seeds.

## 5. Results

## 6. Conclusion

### Declarations

- **Conflicts of Interest:** The authors declare no conflicts of interest.
- **Data availability:** Data will be made available on request to the corresponding authors.

### CRedit authorship contribution statement

**Md. Atikur Rahaman:** Conceptualization, Formal analysis, Methodology, Writing-original draft, Writing-review & editing. **Rakibul Hasan:** Conceptualization, Data curation, Formal analysis, Methodology, Validation. **Md. Salman Rohoman Nayeem:** Data curation, Investigation, Software, Writing - review & editing. **Pratay Paul:** Data curation, Software, Validation, Visualization. **Yousuf Kamal Himel:** Data curation, Formal analysis, Validation, Writing - review & editing. **Mst. Farjana Akter Limu:** Data curation, Software, Validation, Visualization. **Mohammad Nurul Huda:** Investigation, Supervision, Visualization. **Ohidujjaman:** Supervision, Validation, Writing - review & editing.

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